

Dongwoo Kang

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EDUCATION

HANSHIN UNIVERSITY

Bachelor of Engineering in IT Content (College of AI & SW)

Relevant Coursework: Artificial Intelligence, Data Structure, Software Engineering

GPA: 4.11 / 4.5

Osan, South Korea

Feb 2023

EXPERIENCE

KOREA ELECTRONICS TECHNOLOGY INSTITUTE(KETI)

Seong-nam, South Korea

Research on AIoT-Based Lifestyle Pattern Recognition using CSI Signal

May 2024 – Present

- Designed and implemented an end-to-end MQTT-based real-time data pipeline and On-device AI Inference architecture across Linux and Android environments.
- Developed a software-based CSI signal refinement pipeline utilizing Zero-phase Butterworth filters and Singular Value Decomposition (SVD) to effectively eliminate hardware spiking noise and static backgrounds.
- Invented a 'Sequence Time Alignment' algorithm combining a TimeLocked process and linear interpolation to resolve severe packet transmission delays and synchronize data across multiple receivers.
- Designed a lightweight TinyCNN model and established an automated Pytorch to TensorFlow Lite porting pipeline to overcome extreme memory and computational constraints of edge devices.
- Published engineering breakthroughs in multimodal (Vision-CSI) data fusion and multi-receiver synchronization as the first author at prominent international conference, including IEEE ICTC 2024 and IEEE/IEIE ICCE-Asia 2025.

WESANG(YOGIYO)

Seoul, South Korea

Development Support of Merchant Service Platform

Dec 2023 – Feb 2024

- Participated in identifying and resolving minor system issues, supported real-time data management, and assisted in the development of the merchant service platform.
- Modified and tested Python decorators to support internal service logic, while verifying API connectivity between Mobile and backend services to ensure stable system integration.

ACTIVITIES

HCI COLLEGE - COMPANOID INSTITUTE(CIT)

Oct 2024 - Dec 2024

- Defined limitations in existing systems and designed an AI agent-based collaborative system to address context-awareness challenge, including project presentation.
- Conducted user study design and mixed-method UX evaluation to validate hypotheses and inform and industry-oriented business strategy.

PUBLICATIONS

D. Kang, M. Kim and S. Park, "A Study on Fusion Acquisition of Image and CSI Data for Posture Estimation in Indoor Environments," 2024 15th International Conference on Information and Communication Technology Convergence (ICTC)

D. Kang, S. Park and T. Kim, "Robust WiFi Channel State Information-Based Localization via Sequence Time Alignment," 2025 IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia)

HONORS AND AWARDS

Excellence Award of HCI College Conference 2024, Companoid institute

Dec 2024

National Science & Engineering Scholarship, Ministry of Science and ICT

2020 - 2022

Excellence Award of AI & Big Data Hackathon, Hanshin University

Dec, 2021

Academic Excellence Full Scholarship, Hanshin University

2016 - 2020

PROFESSIONAL MEMBERSHIPS

IoT Knowledge Proficiency Test, Korea Association of Smart Homes & IoT	Dec 2025
AWS Certified Solutions Architect - Associate, Amazon Web Services	Oct 2025
SQL Developer (SQLD), Korea Data Agency	Apr 2024
Engineer Information Processing, Q-Net	Apr 2024

TECHNICAL SKILLS

- **Programming** : Python(Proficient), C++, SQL, Swift, Object C
- **AI & Data Science** : PyTorch, OpenCV
- **Tools & Platforms** : Linux (Ubuntu), Git, AWS, Docker, Figma, ComfyUI
- **Database** : MySQL, PostgreSQL

LANGUAGES

- **English** : IELTS Level 6, TEPS 320